



Pharm*Achieve*

ACNE

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

BPO Benzoyl Peroxide

COC Combined Oral Contraceptive

LFTs Liver Function Tests

MI Myocardial Infarction

OTC Over-The-Counter

PCOS Polycystic Ovary Syndrome

Rx Prescription

SMX-TMP Sulfamethoxazole-Trimethoprim

TG Triglycerides

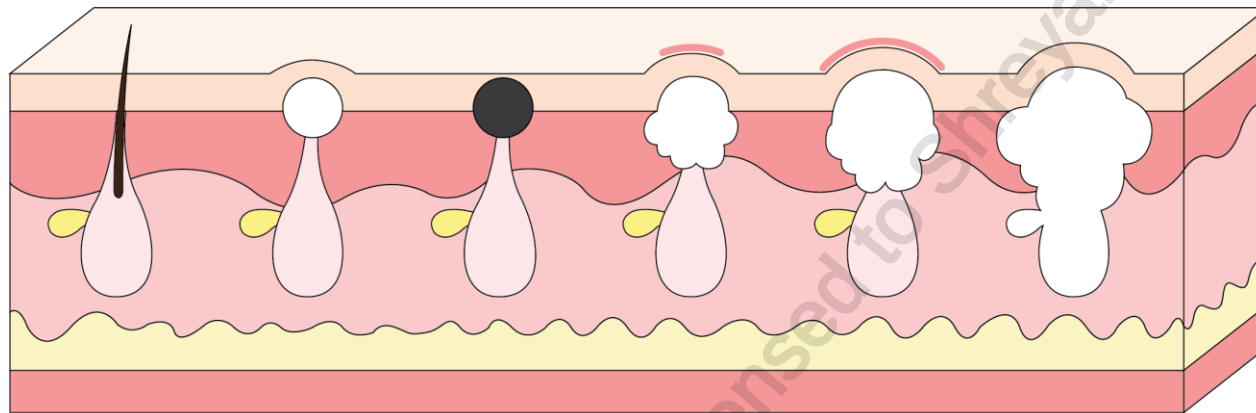
VTE Venous Thromboembolism

OBJECTIVES

- 1 Understand the pathophysiology of acne
- 2 Understand the clinical presentation and distinguish between different severities of acne
- 3 Identify possible risk factors and causes of acne
- 4 Identify red flags for referral
- 5 Discuss available non-pharmacological and pharmacological therapies for acne
- 6 Understand monitoring parameters

DEFINITIONS

- **Acne vulgaris** is one of the most common skin problems
- It is a **multi-factorial inflammatory disease** with **blockage and/or inflammation of pilosebaceous gland** of the skin
- Usually affects young adolescents **from puberty to young adulthood**
- It varies depending on age of onset, distribution, severity, and age of resolution
- It is marked with the presence of comedones, pustules, papules, and/or cysts



Healthy

Whitehead

Blackhead

Papule

Pustule

Cyst/
Nodule

Comedones: Build-up of sebum, bacteria, and dead skin cells in pilosebaceous gland (whitehead, blackhead)

Papules: Raised red bump (without pus)

Pustules: papules containing pus

Cyst: Deep pustule

Nodule: Tender, red, swollen lesions with undefined borders

PATHOPHYSIOLOGY

Excessive sebum production triggered by internal and external factors

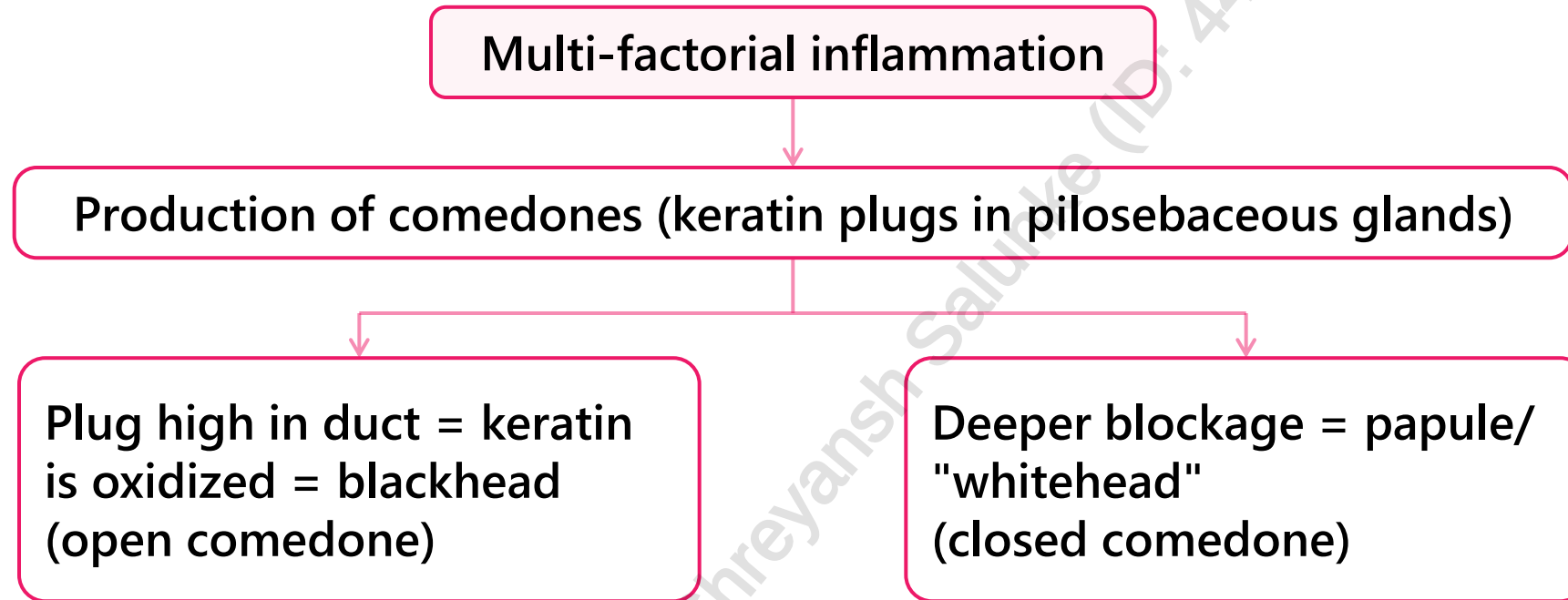
Follicular hyperkeratinization

Proliferation of the anaerobic microorganism
(*Cutibacterium acnes* formerly *Propionibacterium acnes*)

Inflammatory mechanisms

Other: hormones, diet, stress, and genetics also contribute to the pathogenesis of acne

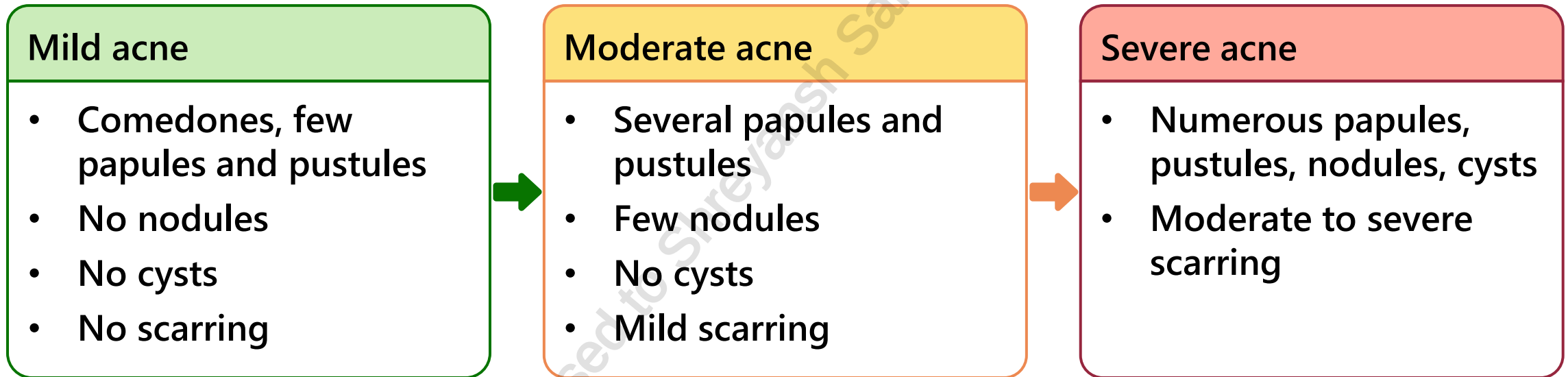
PATHOPHYSIOLOGY



- ✓ Progression leads to cytokine release → inflammatory papule → central pustule
- If inflammatory lesion ruptures below the dermis → cyst or nodule → scarring

CLINICAL PRESENTATION

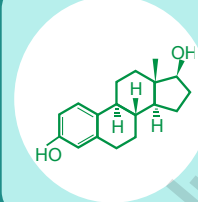
- **Comedonal acne:** open (blackhead) or closed (whitehead)
- **Inflammatory acne:** papules, pustules, nodules, or cysts
- Classification of acne is based on number, type, and distribution of lesions



RISK FACTORS AND CAUSES



Diet (high glycemic load diet, dairy)



Hormones (androgens)



Mechanical/Physical Factors (friction, picking lesions, excessive scrubbing)



Stress



Topical Agents (cosmetic or topical medications)



Drugs (e.g. glucocorticoids, phenytoin, lithium, androgens, progestin-only contraceptives)

DIAGNOSIS

- Based on clinical presentation
- Only to rule out Gram-negative folliculitis: culture and sensitivities of pustules
- **Hormonal investigations** if signs of **hyperandrogenism** including: hirsutism, infertility, irregular periods, insulin resistance, onset in middle-aged women
- **Medication-induced acne**: antiepileptics, antituberculosis drugs, lithium, cobalt in vitamin B12, corticosteroids (consider discontinuation if drug-induced)

WHEN TO REFER

- Presentation at **very young age** (Age <12 years; prior to puberty)
- Onset of age **>30 years**
- **Widespread distributions of lesions**
 - Number of comedones >20, OR
 - Number of inflammatory lesions >15, OR
 - Total number of lesions >30
- **Moderate to severe acne** (presence of nodules or cysts or scarring)
- **Drug or condition-induced** acne
- Non-responsive or worsening symptoms **after 2-3 months of treatment**
- Possible hyperandrogenism (e.g. hirsutism, infertility) or other endocrine abnormality (e.g. PCOS)
- Excessive embarrassment, anxiety, depression, low self-esteem, or feelings of shame



GOALS OF THERAPY

- ✓ Clear and prevent lesions from reoccurring
- ✓ Minimize scarring and dyspigmentation from existing lesions
- ✓ Minimize psychosocial impact

NON-PHARMACOLOGICAL MEASURES

Refrain from picking/squeezing lesions
(delays healing and promotes scarring)

Use gentle skin cleaning techniques
(e.g. avoid rough scrubbing; use oil-free,
alcohol-free, non-comedogenic products
and remove makeup before bed)

Eat a low glycemic diet and limit dairy
intake
(uncertain evidence)

Avoid using occlusive products on affected
areas, minimize stress* (controversial)

Chemical peels, microdermabrasion, and
photodynamic therapy have little evidence
supporting efficacy

MILD ACNE TREATMENT ALGORITHM

BPO
OR topical retinoids
(tretinoin/adapalene/ tazarotene)
OR both
OR BPO/clindamycin
OR BPO/adapalene

If no improvement
after 2-3 months

Combined oral contraceptive

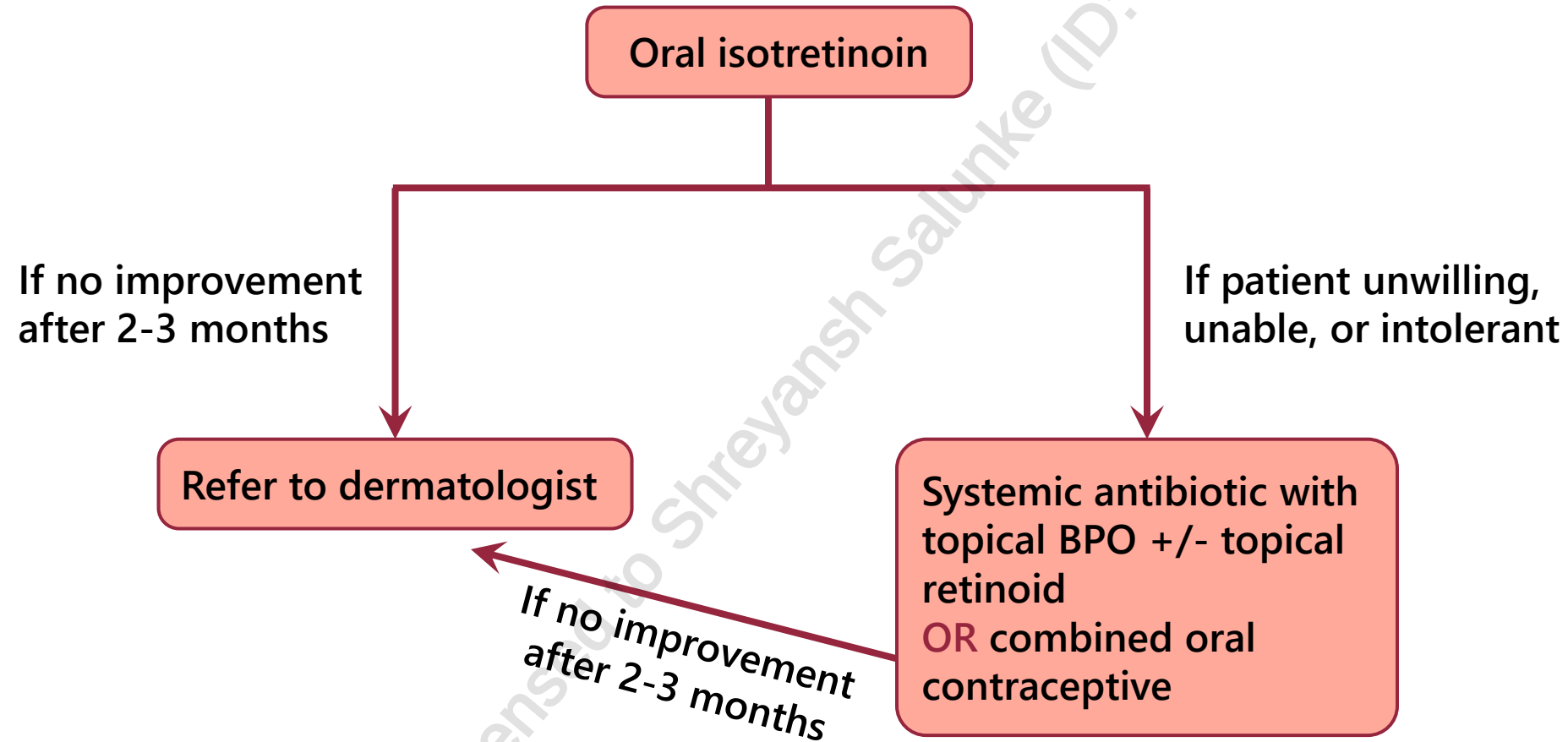
MODERATE ACNE TREATMENT ALGORITHM

BPO
OR topical retinoids
OR both
OR BPO/clindamycin
OR BPO/adapalene
+
COC or systemic antibiotic
(tetracyclines)

If no improvement
after 2-3 months

Refer to dermatologist

SEVERE ACNE TREATMENT ALGORITHM



PHARMACOLOGICAL MEASURES

TYPES OF TOPICAL VEHICLES

Oily skin

Solutions/gels



Dry or sensitive skin

Creams



Any skin type

Lotions



PHARMACOLOGICAL MEASURES

OVERVIEW

Topical Agents	• Exfoliants: Glycolic acid, salicylic acid, sulfur (OTC) • Antimicrobials with Anti-inflammatory properties: Benzoyl peroxide (OTC/Rx), dapsone • Antimicrobials with Keratolytic properties: Azelaic acid • Retinoids: Adapalene, tazarotene, tretinoin, trifarotene • Antibiotics: Clindamycin, erythromycin • Newer therapies: → Winlevi® (clascoterone) → Cabtreo® (clindamycin/adapalene/benzoyl peroxide)
Oral Agents	• Antibiotics: tetracycline, doxycycline, minocycline, trimethoprim-sulfamethoxazole, azithromycin • Hormonal therapy: COCs, spironolactone • Retinoids: isotretinoin

PHARMACOLOGICAL MEASURES

NON-PRESCRIPTION

Benzoyl peroxide (BPO) anti-microbial and anti-inflammatory

- 1st line for comedonal or mild acne
- No prescription required for concentrations $\leq 5\%$

Glycolic acid exfoliant

Salicylic acid exfoliant

- Less effective than benzoyl peroxide

Sulfur exfoliant

- Often combined with salicylic acid

PHARMACOLOGICAL MEASURES

TOPICAL AGENTS

Benzoyl peroxide (BPO): anti-microbial and anti-inflammatory

- **Antibacterial and mildly comedolytic**
- **Apply once daily**, gradually increase to 2-3 times/day prn
- May take up to **3 months** to see improvement [initial worsening can occur in first 2 to 4wks]
- Can prevent and eliminate *C. acnes* → no evidence of bacterial resistance
- **2.5%** shows similar efficacy to 5% and 10% (need prescription for >5% strengths)
- Gel formulation improves absorption
- Degrades topical retinoids → **must separate**
- Can **bleach** clothing & hair
- **Adverse effects: bleaching, skin irritation, peeling**
Skin irritation can be decreased by using lower concentrations, reduce the amount of time the drug is left on the skin, and use water-based preparation over alcohol-based

Salicylic acid (1-2%): exfoliant

- **Comedolytic**
- Concentrations higher than 1-2% are usually more irritating
- Not as effective as topical retinoids

Glycolic acid (2-15%): exfoliant

- **Mildly comedolytic**
- Used when topical retinoids not tolerated

PHARMACOLOGICAL MEASURES

TOPICAL AGENTS

Azelaic acid (15%): antimicrobial and keratolytic

- **Antibacterial and comedolytic**
- Apply **twice daily**
- Does not promote resistance
- Some effect at reducing inflammation but not as much as BPO
- Most effective in combination with other agents
- Can cause **hyper- or hypo-pigmentation** (use with caution in patients with dark skin complexions)

Dapsone: anti-inflammatory

- Apply **twice daily**
- Best used in inflammatory acne
- Mildly irritating and available in a topical gel formulation
- **Not used orally** due to serious side effects e.g. agranulocytosis

Clindamycin, erythromycin: antimicrobials

- Apply **once or twice a day** depending on formulation
- Slow acting, for papulopustular acne to decrease *C. acnes*
- Resistance often develops over time with monotherapy → **use with benzoyl peroxide** to decrease chance of bacterial resistance
- May take **8-12 weeks** to see improvement

PHARMACOLOGICAL MEASURES

TOPICAL AGENTS

Intralesional topical corticosteroids (triamcinolone)

- For occasional or stubborn nodule or cyst (not used on multiple lesions)
- Fast-acting to effect and reduces pain; injection side effects present

Topical retinoids/retinoid analogues (vitamin A)

Strongest comedolytic agents (inflammatory & non-inflammatory lesions)

- **Apply at bedtime** and separated from benzoyl peroxide (AM)
- Efficacy may take **months**
- Includes **tretinoin** (Retin-A®), **adapalene** (Differin®), **tazarotene** (Tazorac®), **trifarotene** (Aklief®)
- Tretinoin (most cost-effective but most photosensitizing), adapalene (least irritating), tazarotene (most potent)
- Micronized formulation of tretinoin → improves delivery of retinoid and reduces irritation
- **Adverse effects: skin dryness, redness, skin peeling** (improves over time and spots may get worse before they improve; use a good moisturizer to prevent skin dryness)
- **Skin is more sensitive to sunlight**; drug degrades with sun exposure (use sunscreen)
- **Contraindicated in pregnancy**

PHARMACOLOGICAL MEASURES

SYSTEMIC AGENTS

Systemic Antibiotics (tetracyclines)

- Tetracyclines are considered **1st-line** (doxycycline, tetracycline > minocycline)
 - Minocycline is associated with more side effects: drug-induced lupus, hepatitis, cutaneous pigmentation, dizziness, vertigo
- Consider in **moderate-severe acne** or if **topical therapy is ineffective after 2-3 months**
- Trial 6 weeks of therapy for effectiveness, limit duration to 3 months, use with topicals
- **Combine with BPO to ↓ resistance** → DO NOT use as monotherapy due to risk of bacterial resistance
- Reserve other antibiotics (macrolides, SMX-TMP) for last line
- **Adverse effects:** GI upset, yeast overgrowth, photosensitivity, pseudotumor cerebri (rare)
- **Contraindicated in pregnancy and in children <8 years of age**

PHARMACOLOGICAL MEASURES

SYSTEMIC AGENTS

Oral retinoids (isotretinoin, Accutane®)

- **Most powerful anti-acne agent**, ↓ sebum production & ↓ inflammation
- Useful for severe nodulocystic & non-responsive acne
- Most effective treatment with remission lasting beyond duration of therapy
- Teratogenic: contraindicated in pregnancy & breastfeeding
- **Women of childbearing age**: complete baseline & monthly pregnancy tests & 2 reliable methods of contraception **until 1 month after last dose**
- Wait 8 weeks after therapy completion before pursuing additional courses
- Contraindicated with tetracyclines due to risk of pseudotumor cerebri. *What if a patient is switching from a tetracycline antibiotic and wants to start on isotretinoin?*



Considered
1st line for
severe acne

Common Adverse Effects

- Drying/chapping of skin (moisturize)
- Headache
- Myalgias
- Arthralgias
- Photosensitivity (use sunscreen)
- Mild nosebleed

Uncommon/ Rare Adverse Effects

- Lipid abnormalities (hypertriglyceridemia)
- ↑ in LFTs
- Pseudotumor cerebri
- Mood disorders (depression, suicidal ideation)
- Erythema multiforme
- SJS/TEN

PHARMACOLOGICAL THERAPY

SYSTEMIC AGENTS

Hormonal Therapy

COMBINED ORAL CONTRACEPTIVES (COCs)

- If **acne worsens with menstrual cycles**: Consider COCs
- COCs with minimal androgenic effects or with antiandrogenic effects: cyproterone acetate or drospirenone (↑ VTE risk)
- **Discontinue 3-4 cycles after acne has resolved**
- Adverse effects: breakthrough bleeding, nausea/vomiting, weight gain, bloating, breast tenderness, depression, headaches, chloasma
- Rare adverse effects: thromboembolism, stroke, MI, retinal artery thrombosis, benign liver tumour, gallstone formation, hypertension
- **DI**: Rifampin consistently decreases estrogen levels (↓ effectiveness)

SPIRONOLACTONE

- Form of **antiandrogenic hormonal therapy** used in selected women (expert opinion)
- **Effective contraception recommended** in women of child-bearing age
- **Adverse effects**: Hyperkalemia, diuresis, irregular menses, breast tenderness, nausea, headache, fatigue, dizziness

SPECIAL POPULATIONS

PREGNANCY

✓ Recommended

- Benzoyl peroxide
- *Erythromycin (systemic and topical)
- Topical clindamycin
- Glycolic acid and azelaic acid
- Intralesional corticosteroids (triamcinolone)



✗ Not Recommended

- Isotretinoin (systemic)
- Retinoids (systemic and topical)
- *Erythromycin (estolate salt)
- Sulfamethoxazole/trimethoprim (SMX-TMP)
- Spironolactone
- Tetracycline, doxycycline, minocycline (especially in 2nd/3rd trimester)
- Oral contraceptives

SPECIAL POPULATIONS

BREASTFEEDING

✓ Generally Compatible with breastfeeding

- Topical antibiotics
- Benzoyl peroxide
- Topical retinoids
- Azelaic acid
- Glycolic acid
- Spironolactone
- Erythromycin (increased risk of pyloric stenosis)
- Tetracycline (<3 weeks)



✗ Not Recommended

- Retinoids (systemic)
- Isotretinoin (systemic)
- Sulfamethoxazole-trimethoprim (SMX-TMP)
- Tetracycline, doxycycline, minocycline

PATIENT COUNSELLING

Wash face gently with tepid water and mild antiseptic cleanser

Avoid abrasive cleansers, vigorous scrubbing, and astringents

Apply a sunscreen daily and avoid excessive sun exposure

Use a non-comedogenic moisturizer to reduce irritation

Apply thin layer of cream

MONITORING

- Assess efficacy of treatment at **2 to 3 months**
- If there is **no response or symptoms are worsening** after 2 to 3 months → **REFER**
- If there is an **inadequate response**:
 - Consider switching topical agents or combining agents
 - Consider adding a topical antibiotic for inflammatory symptoms
- If the patient **responds well** to treatment, may contact primary care provider for refills
 - Benzoyl peroxide and topical retinoids can be used for maintenance
- **Assess for adverse effects** and recommend management
 - Skin dryness: fragrance-free, water-based moisturizing cream
 - Excessive skin irritation with benzoyl peroxide/retinoids: ↓ strength & frequency

MONITORING PARAMETERS

EFFICACY ENDPOINTS		
Parameter	Degree of Change	Timeframe
Acne presentation (e.g. comedones, papules, pustules)	Improved	in 2-3 months

SAFETY ENDPOINTS		
Parameter	Degree of Change	Timeframe
ADRs from topical therapy (e.g. skin dryness, excessive irritation)	Prevent or minimize	Duration of therapy
ADRs from oral therapies		
With isotretinoin: LFTs and lipid profile (specifically TG)	None (within normal levels)	Measured at baseline before treatment and at 8 weeks after initiation

THERAPEUTIC TIPS

Topical therapy should be applied as frequently as prescribed to the affected zones and not to the lesions alone

Cream is better for patients with dry/ sensitive skin while gels are best for oily skin or those who live in hot and humid climates

When used together, apply BPO in the AM and retinoid in the PM

Topical benzoyl peroxide/antibiotic formulations have short shelf-lives

Gradually taper up tretinoin to avoid local irritation and increase tolerability (e.g. less frequent applications, shorter contact times)

Use BPO with topical antibiotics to decrease the chance of bacterial resistance

Allow for several weeks of treatment to assess effectiveness

Acne is a disease which could last years, once control is achieved regimen can be simplified

A 7-day wash-out period when switching from a tetracycline antibiotic to isotretinoin

TAKEAWAY

Mild acne

- Topical medications:
 - BPO, retinoids, BPO + retinoids, OR fixed-dose combinations of BPO with adapalene/clindamycin
 - If treatment failure with above, consider clindamycin/tretinoin gel
- Those with dry, sensitive skin = creams or lotions are preferred (less drying)
- Those with oily skin = gels (less greasy)
- If topical therapies are ineffective after 2-to-3-month trial, then add COCs or systemic antibiotics

Extensive moderate papulopustular acne

- Systemic antibiotics + topical medications
- Women can use oral contraceptives + topical medications

Severe acne

- Oral isotretinoin
- If isotretinoin intolerant/contraindicated, use systemic antibiotics + BPO +/- topical retinoids OR COCs

CASE

JT is an 18-year-old female presenting to the pharmacy with concerns about her recent acne breakout. JT explains that she has experienced episodes quite frequently in the last year, but it has not bothered her until now. She will be attending university soon and wants her acne to be cleared by then. She has not tried anything in the past to reduce her symptoms and is looking for your advice on selecting a product. JT has no significant medical history. She is not pregnant or breastfeeding. She works a part-time job in a factory. She currently takes cetirizine 10 mg daily PRN for seasonal allergies (pollen).

1. JT has mild acne. Which of the following is true regarding acne treatment?
 - a) Improvement of symptoms is generally seen within 2 weeks of initiating topical therapy
 - b) Combination therapy of multiple topical agents is not recommended for mild acne
 - c) It is best to apply benzoyl peroxide 5 minutes after the application of topical adapalene
 - d) Benzoyl peroxide 2.5% is as effective as 5% and 10% strengths and is less irritating



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2. Which of the following is NOT a non-pharmacological measure to recommend to JT?
- a) Refrain from picking or squeezing lesions
 - b) Use an oil-free, alcohol-free, comedogenic facial wash
 - c) Avoid use of occlusive products on affected area
 - d) Avoid hot and humid climates if possible



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3. Which of the following is NOT a counselling point if JT were to initiate topical benzoyl peroxide?
- a) Benzoyl peroxide could stain clothing or bleach hair
 - b) If dryness/peeling occurs, decrease the frequency or concentration
 - c) Combine with a topical antibiotic to reduce the incidence of bacterial resistance
 - d) Water-based formulations can help limit dryness and peeling



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4. JT is now a 29-year-old patient who presents to her doctor with papules and pustules all over her face as well as nodules and cysts. Which of the following options would be most suitable?
- a) Benzoyl peroxide
 - b) Doxycycline + benzoyl peroxide
 - c) Doxycycline + ethinyl estradiol/cyproterone acetate
 - d) Isotretinoin



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5. Keow S, Xiong G, Abu-Hilal M. Update to acne vulgaris treatment for Canadian practice. *Can Fam Physician*. 2025;71(7-8):455-466. doi:10.46747/cfp.710708455.

CHANGE LOG

March 2025

- Minor formatting changes made throughout
- Content reviewed; no clinical changes made

August 2025

- Slide 16: Added an arrow to algorithm of severe acne treatment
- Slide 18: Added Cabtreo® – a new combination treatment recently approved in Canada
- Slide 36: Updated references with new CFP review article

October 2025

- Formatting changes; some slides re-arranged
- Slide 11: Addition of scarring for reasons for referral for severe acne
- Slide 18: Added Winlevi® to topical therapies
- Slide 31: Addition of tip when using BPO and retinoid together, addition of using BPO with topical antibiotics, addition of washout period

February 2026

- Slide 24: Added increased LFTs to side effects of isotretinoin
- Slide 30: Added monitoring parameters slide